

The Measure Makes the Finding

How measurement, design, and framing shape what we conclude about race, democracy, and punishment in the U.S. states

Miura Meng

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Abstract

Conclusions about racial inequality often depend more on how we measure and frame a question than on the evidence itself. That measurement and framing are not neutral is by now well established; what is rarer is a compact, concrete demonstration of how far the answer can move, and how easily the moves compound, when a single high-stakes question is carried through the ordinary choices an analyst makes. This paper offers such a demonstration, using race, democracy, and punishment across the U.S. states. It walks the case through four routine decisions. The choice of inequality measure can make more democratic states look either more or less racially unequal, because the popular Black/White ratio moves with its White denominator, which falls steeply in more democratic states while Black imprisonment changes much less. The choice between comparing states and following one state over time turns a striking cross-sectional pattern into a small within-state one. A clustering method returns a tidy five-group typology that looks more solid than the data warrant. And Florida's 2018 Amendment 4 reads as a sweeping expansion of voting rights or as a non-event, depending on whether one reads its text or its effect; a follow-on law (SB7066) requiring payment of court fines left turnout unmoved. In each case a small decision settles the answer. The substantive findings are secondary to that methodological point. A closing analysis shows the demonstrations need not stay separate: one Bayesian multilevel model of the underlying rates reproduces the competing measures as contrasts of a single posterior, and reports the reversal as a probability. Methods include multilevel models, group-based trajectory modeling, and synthetic control.

Introduction

A finding rarely survives retelling intact. As a result moves from the analysis to the abstract to the headline, its conditions fall away, and a claim that began as tentative comes to sound certain. The same number, framed one way, points to individual responsibility; framed another way, it indicts institutions, even though nothing underneath has changed. Quantitative social science is especially exposed to this, because a single summary figure can quietly carry all the choices that produced it.

This paper takes that vulnerability as its subject. On the surface it asks a substantive question: across the U.S. states, does a healthier democracy go together with less racial inequality in punishment, and can restoring the vote to people with felony convictions change anything? Underneath, though, the argument is methodological, because at every step the answer turns on a decision that is easy to make without noticing. Which measure of inequality should be reported? Should states be compared with one another, or each state with itself over time? How many groups should be read into the data? And is a reform “large” by its text or by its effect? Each of these choices, it turns out, can flip the conclusion.

The paper proceeds in four demonstrations, and each one revises the one before it.

- The first measures racial inequality across states. By one common measure, more democratic states look more unequal; by three others, they look the same or better.
- The second compares each state with itself over time, and the cross-state result largely dissolves.
- The third sorts states into trajectory types. The types are real enough to describe, but the method reads in more structure than the data can support.

- The fourth treats Florida’s 2018 voting-rights reform as a natural experiment. By its text it freed 1.4 million voters; by its effect it changed almost nothing.

The point is not any single number, but the pattern across them: measurement, design, and framing decide what we are able to conclude, and careful choices keep overturning the answer that looked settled. The case, race and democracy and punishment, matters in its own right, yet here it mainly serves to make that methodological lesson concrete.

Background

Measurement, framing, and what counts as a finding

A claim about inequality is never simply a reading of the evidence. It is also a product of the choices made in turning evidence into a number, and a number into a sentence. Scholars writing under banners such as data feminism and QuantCrit have pressed this point in recent years, arguing that data are not neutral and that statistics, far from speaking for themselves, carry the assumptions of whoever collected, classified, and summarized them (D’Ignazio & Klein, 2020; Gillborn, Warmington, & Demack, 2018). The complaint is not that quantification is worthless but that it hides its own premises. Measurement of inequality is a clear example: faced with the same data, an analyst can often choose the measure that best supports a preferred conclusion, and the decision between an absolute and a relative measure alone can send a trend in opposite directions (Kjellsson, Gerdtham, & Petrie, 2015; Moonesinghe & Beckles, 2015). Framing does comparable work even when the numbers are held fixed. Describing a gap as one of “achievement” rather than “opportunity,” for instance, shifts where readers locate its causes, away from structures and toward individuals, with-

out altering a single statistic (Quinn, 2025). And as a result travels outward from the model that produced it, the qualifications that gave it meaning tend to drop away, so that a conditional finding begins to circulate as a settled fact (van der Bles, van der Linden, Freeman, & Spiegelhalter, 2020; Merry, 2016).

As a principle, none of this is new. What it has lacked is a compact, concrete demonstration: a single consequential question carried through the ordinary choices an analyst makes, with the answer shown to move at each step. That is what this paper sets out to provide. It takes one case and walks it through four routine decisions, treating each decision as part of the analysis rather than as a preliminary to it. The contribution, then, is not the insight that measurement and framing matter, which is by now well established, but the worked example that shows how much they matter, and how readily they compound, once a real claim about racial inequality is at stake.

Measuring racial disparity in incarceration

The most common summary of racial disparity in imprisonment is a single ratio, the Black imprisonment rate divided by the white rate. Its appeal is obvious, but it is a quotient, and a quotient can change because its numerator changes, because its denominator changes, or because of both at once. A state can post a high ratio because it imprisons Black residents at an unusually high rate, or because it imprisons white residents at an unusually low one, and these are very different social facts that the ratio reports identically. Criminologists have long shown that conclusions about racial disproportionality hinge on such choices: on how much of a gap is attributed to differential involvement rather than to downstream processing (Beck & Blumstein, 2018), on which benchmark population is placed in the denominator (Cesario, Johnson, & Terrill, 2019), and even on the

data source and racial-classification scheme, which can widen or narrow the measured Black/White gap (Sabol, Johnson, & Lynch, 2025). Rather than commit to one measure in advance, this paper carries three in parallel, the ratio, the absolute Black rate, and the gap between the Black and white rates, and treats their disagreement as part of the evidence.

The case: state democracy, disenfranchisement, and Florida

State democracy can be measured and tracked. The State Democracy Index scores each state on features such as electoral participation, the fairness of district maps, and the accessibility of voting, and those scores move over time, including downward in a subset of states (Grumbach, 2022, 2023). Criminal disenfranchisement, meanwhile, is one of the oldest and most racially loaded boundaries on the vote. Many states wrote or expanded these laws in the decades after the Civil War, some in terms understood at the time as a way to limit Black political power, and the legacy persists: roughly four million Americans remain barred from voting by a felony record, with Black citizens barred at several times the rate of others (Manza & Uggen, 2006; The Sentencing Project, 2024). Florida concentrated this history into a single recent episode. In 2018 its voters passed Amendment 4, restoring the vote to about 1.4 million people upon completion of their sentences. Within months, however, the legislature passed SB7066, which redefined “completion” to require paying all outstanding court fines and fees first, a condition critics likened to a poll tax and a federal appeals court allowed to stand in 2020.

Data and measures

The analysis combines three public datasets, each at the state-year level. The State Democracy Index 2.0 (Grumbach & Bitton, 2024) scores the fifty states from 2000 to 2023, with higher values meaning more democratic. The Vera Institute’s Incarceration Trends provides race-specific imprisonment rates per 100,000; the Black and white rates used here run from 1990 to 2022. The UF Election Lab turnout series (McDonald, 2023) gives voting-age-population (VAP) turnout for each even-year election, along with the number of people disenfranchised by a felony record.

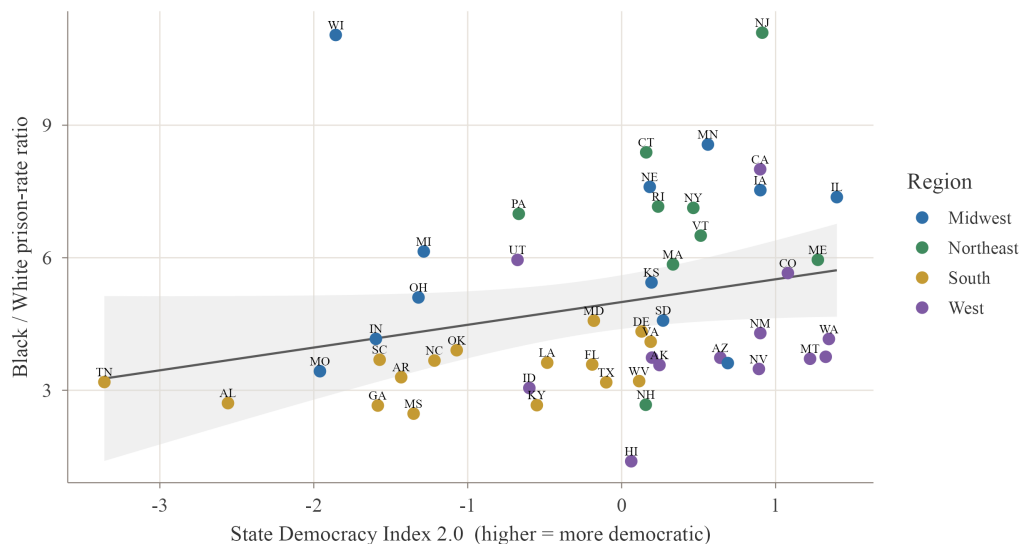
Because racial inequality has no single definition, the paper follows three measures side by side: the Black/White rate ratio, the absolute Black rate, and the gap between the Black and white rates. Over this period the national ratio sits near a median of 5.4. For the natural experiment the outcome is VAP turnout, whose denominator is the voting-age population and so does not move mechanically when people are re-enfranchised, and the first-stage quantity is the share of the voting-age population disenfranchised by a felony record.

Demonstration 1: the measure decides the sign

Begin with the question in its most direct form. For each state I average the Black/White imprisonment ratio and the State Democracy Index over 2016 to 2020, and plot one against the other (Figure 1). The line slopes upward: the more democratic a state is, the wider its Black/White ratio tends to be. Read at face value, this is an uncomfortable result, because it seems to say that democratic health and racial fairness in punishment are in tension, and that the states doing democracy best are the ones treating Black and white residents most unequally behind bars.

Weaker state democracy, wider racial incarceration gap?

Each point = one U.S. state, averaged 2016-2020. y = Black-to-White prison-rate ratio.



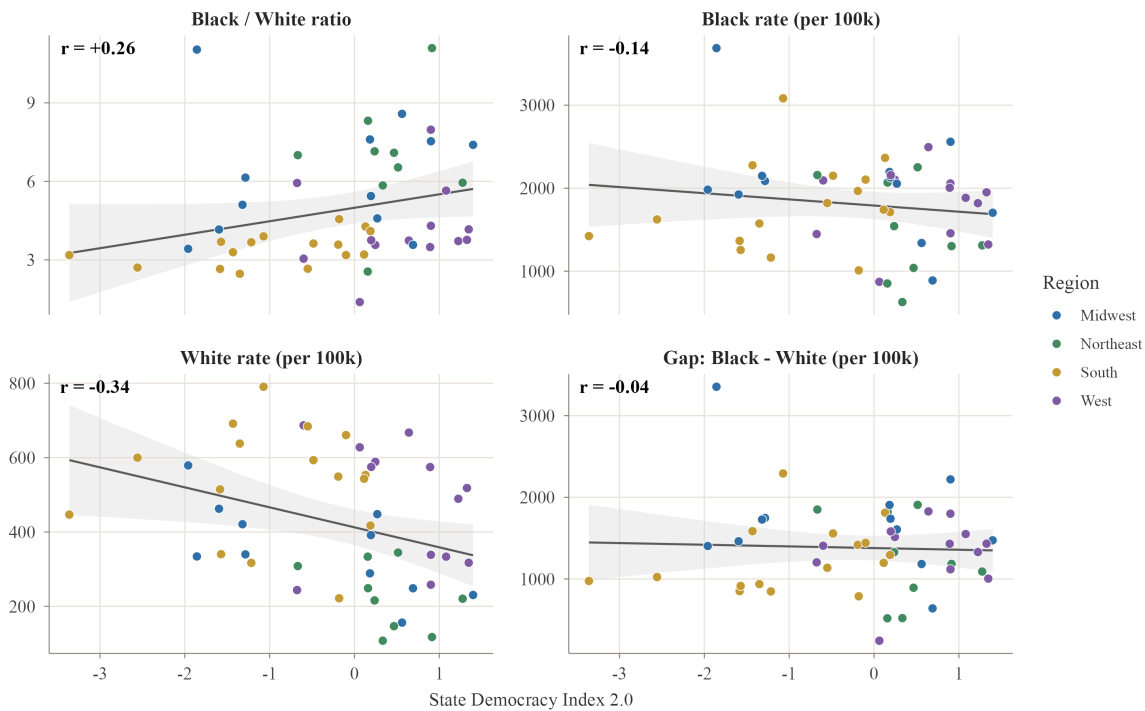
Sources: Grumbach & Bitton, State Democracy Index 2.0 (UC Berkeley); Vera Institute, Incarceration Trends.
Descriptive association only - not causal.

Figure 1: Across states, averaged over 2016 to 2020, the Black/White incarceration ratio rises with the State Democracy Index.

Before accepting that, it is worth asking whether the result is a fact about the world or a fact about the ratio. The test is simple. I hold the horizontal axis fixed, so the measure of democracy never changes, and I substitute different measures of “racial inequality” on the vertical axis (Figure 2). If the relationship is real, it should survive the substitution. It does not. The Black/White ratio correlates positively with democracy, at about +0.26, which is the pattern we just saw. The absolute Black imprisonment rate, though, correlates slightly negatively, at about -0.14 , so by that measure more democratic states imprison Black residents at marginally lower rates rather than higher ones. The gap between the Black and white rates is essentially flat, at about -0.04 . And the absolute white rate correlates strongly negatively, at about -0.34 . Four measures of the supposedly same inequality, and they do not merely differ in strength; they point in different directions.

One democracy axis, four ways to measure 'racial inequity' - the sign flips

Each point = one U.S. state, averaged 2016-2020. x = State Democracy Index 2.0 (higher = more democratic).



Sources: Grumbach & Bitton SDI 2.0; Vera Institute Incarceration Trends. Descriptive / cross-sectional only.

Figure 2: The same democracy axis against four measures of racial inequity. The sign of the relationship flips depending on the measure.

The reason comes into focus once we look at what the ratio is made of. It is the Black rate divided by the white rate, so its value depends as much on the denominator as on the numerator. More democratic states, it turns out, imprison white residents at strikingly low rates, and that small denominator does the arithmetic work, inflating the ratio even in a state where Black imprisonment is no higher than average. The South illustrates the same mechanism in reverse. Southern states post lower ratios, but not because they treat Black and white residents more evenly; they imprison white residents heavily too, and the large white denominator pulls the ratio down. The regional averages put numbers on this. White imprisonment runs about 230 per 100,000 in the more democratic Northeast and about 535 in the South, more than twice as high, while the Black rate stays

within a much narrower band, roughly 1,460 to 2,060 across the four regions. The ratio therefore runs highest on average across the low-white-rate Northeast and lowest across the high-white-rate South, even though the South is the less democratic region. The measure reported most often, then, is also the one most exposed to movement that has nothing to do with how Black people are treated, since a state can climb the ratio simply by incarcerating fewer white people. Judged by the absolute Black rate, more democratic states look slightly better; judged by the gap, they look no different; judged by the ratio, they look worse. Nothing about the data has changed between those three sentences. Only the measure has.

This is not a quirk of incarceration data. The divergence between absolute and relative measures of disparity is a known formal result, demonstrated in health research where the two can move in opposite directions over the same period (Moonesinghe & Beckles, 2015), and the dependence of measured racial disparity on such choices has been shown directly for imprisonment (Sabol, Johnson, & Lynch, 2025). What the figure adds is a vivid sense of how cleanly the choice can reverse a headline: one decision, made before any modeling begins, has already settled whether democracy looks like a friend or an enemy of racial equality.

Demonstration 2: comparing states, or comparing a state with itself

Comparing different states with one another folds democracy together with everything else that distinguishes them, including region, demographics, history, and the local economy, and the first demonstration has already shown how badly that folding can distort an answer. A cleaner question

is longitudinal. When a single state's democracy rises or falls, does its incarceration move with it? Posed this way, each state serves as its own control, and the fixed differences between states drop out of the comparison.

To carry this out I split the democracy score into two parts. The first is each state's long-run average, which holds the between-state variation that the earlier figures relied on; the second is each year's deviation from that average, which holds the within-state variation that tracks change over time. The split follows Mundlak (1978). I then fit a linear mixed model with a random intercept for each state and a common year trend, using the lme4 package (Bates et al., 2015), so that the two kinds of variation are estimated side by side rather than confused with each other (Figure 3).

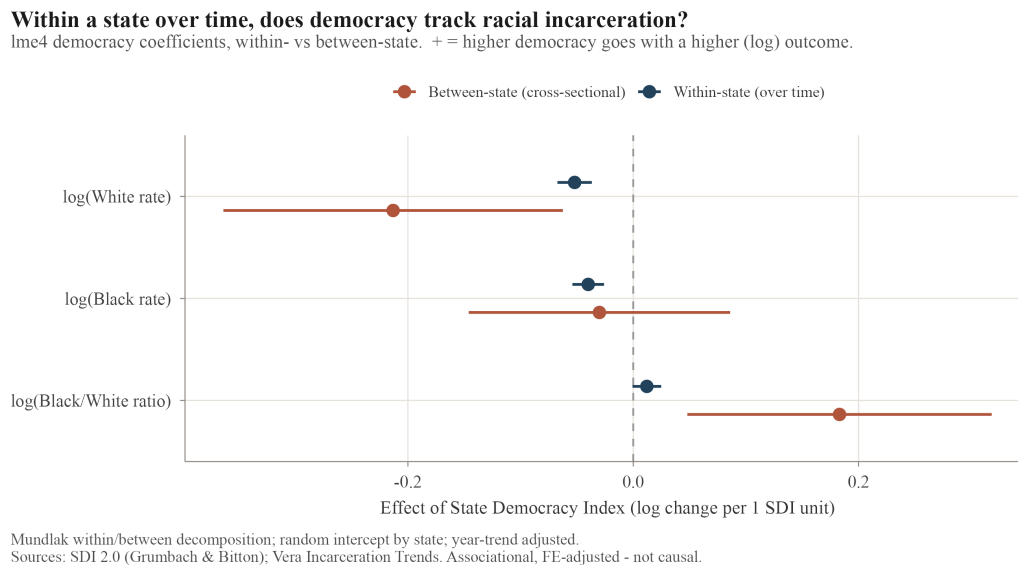


Figure 3: Within-state versus between-state democracy coefficients. The cross-sectional ratio effect (orange) disappears within states (blue).

The contrast is clean. The alarming cross-sectional result for the ratio turns out to be entirely between-state, at about +0.18, and to vanish within states, at about +0.01. In other words, it reflects which states are which, not what happens when a given state's democracy changes. Within states,

stronger democracy goes with lower incarceration of both groups, by roughly -0.04 for the Black rate and -0.05 for the white rate in log points per index point, and because the two rates fall together the ratio barely moves, exactly as the first demonstration would predict. The largest term in every model, however, is neither within nor between democracy but the year trend: the ratio fell about 2.6 percent a year and the Black rate about 2.7 percent, a nationwide decline that has little to do with state democracy. An analysis that left time out would mistake that tide for its own discovery. So the within-state link is real and consistent in direction, but it is modest, and the paper is stronger for saying so. One choice of comparison, cross-section against within-state, has already turned a striking effect into a small one.

From multiverse to model

The title marks a move. The four demonstrations are a small multiverse, one question carried through choices that should not change the answer so that the disagreements among the answers can show; this section turns that multiverse into a single Bayesian model, in which the same choices reappear as contrasts of one posterior. The multiverse is the better tool for revealing that the choices matter; the model is the better tool for saying how much, and how reliably.

The first two demonstrations treated the choice of measure and the choice of comparison as separate forks in the analysis. They are not separate. The four measures and the within and between contrasts are all summaries of the same two underlying quantities: how heavily a state imprisons its Black residents, how heavily it imprisons its white residents, and how each of those moves with democracy. Rather than walk the forks one at a time, then, we can fit a single model of those two quantities and read every fork off the one set of estimates it implies.

To do this I fit one joint multilevel model with two outcomes at once, the log Black imprisonment rate and the log white rate, each allowed to depend on democracy. Democracy enters in the Mundlak form of the second demonstration, split into a between-state part, a state's long-run average, and a within-state part, its year-to-year deviation; each state keeps its own intercept, and the two outcomes' intercepts and residuals are free to correlate. The model is Bayesian, fit with weakly informative priors using the brms package (Bürkner, 2017), and it converged cleanly, with every R-hat below 1.01 and no divergent transitions. The coefficients below are scaled per one standard deviation of democracy rather than per index point, so their sizes are not directly comparable to those in the second demonstration, but their signs and pattern are the same.

Once the model is fit, the four measures are no longer four analyses but four contrasts of one posterior (Figure 4). Between states, a one standard deviation increase in democracy goes with white imprisonment lower by about 0.17 log points (95 percent credible interval -0.29 to -0.05) and Black imprisonment that is essentially flat (-0.02 , -0.12 to $+0.07$). The ratio effect is just the difference between these two, and it comes out positive, about $+0.14$ ($+0.03$ to $+0.25$), for exactly the reason the first demonstration gave: the ratio rises not because Black imprisonment rises but because the white denominator falls. The alarming cross-sectional pattern and the reassuring one are the same posterior, read two ways. Within states the two rates fall together, by about -0.02 for the Black rate and -0.03 for the white rate, with both intervals well below zero, and the ratio barely moves, $+0.01$ with an interval that straddles zero, just as the second demonstration found.

Four 'measures' are four contrasts of one posterior

Effect of +1 SD between-state democracy. The ratio is positive only because white imprisonment (blue) falls faster than Black: the denominator does the work.

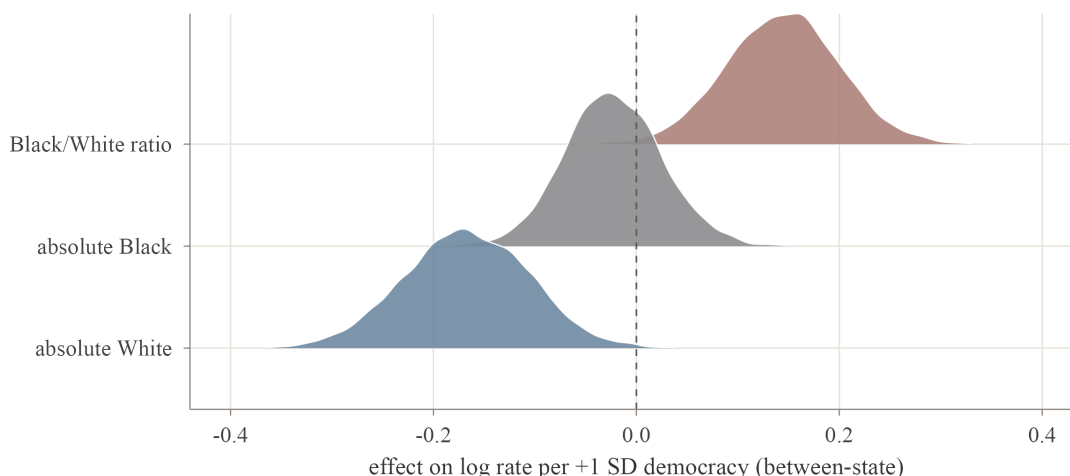


Figure 4: The four measures of racial inequality as contrasts of one posterior. Each density is the effect of a one standard deviation increase in between-state democracy on one quantity. The Black/White ratio (red) sits above zero only because white imprisonment (blue) falls faster than the Black rate (grey); a single posterior produces a rising ratio and a falling white rate at the same time.

Holding the measures inside one model buys something the side-by-side comparison cannot. Because every contrast is computed from the same draws, we can ask not just whether the measures disagree but how reliably they do. The posterior probability that democracy raises the ratio while leaving white imprisonment lower, which is the exact shape of the first demonstration's reversal, is about 0.99. The weaker version, that the ratio rises while the absolute Black rate falls, holds with probability about 0.68, lower only because the Black rate sits close enough to flat that its sign is uncertain. The reversal is therefore not a fragile accident of one arbitrary comparison. It is a near-certain feature of how the two rates move, and the model lets us report it as a probability rather than as a juxtaposition of figures.

The pattern is not an artifact of the likelihood or of region. Replacing the Gaussian model on the

log rates with a negative-binomial count model on the raw counts leaves the estimates essentially unchanged, and adjusting for census region as a fixed effect shrinks the between-state ratio effect only modestly, from about +0.14 to +0.11, while the falling white rate and the near-certain reversal both hold.

None of this makes the relationship causal. The model rearranges the same observational evidence more economically; it does not identify an effect. Its value is methodological. The earlier demonstrations show, in the accessible form of a multiverse, that the answer moves with the analyst's choices; this model shows why, by placing those choices as contrasts of a single set of estimates, and it puts a number on how much each choice matters. A multiverse is the better way to reveal the problem, and a model that absorbs the sources of variation is the more disciplined way to study it (Gelman et al., 2012). The two are complementary, and the counsel to carry more than one measure only gains force once the measures can be shown to be one model seen from different sides.

Demonstration 3: a method that reads in more than the data hold

An average describes the typical state but conceals the differences between states, and those differences are part of the story. So the next question is whether states fall into recognizable types according to how democracy and incarceration moved together. Group-based trajectory modeling, developed in criminology by Nagin and Land (1993) to classify offending over the life course, is built for exactly this. Using the gbmt package (Magrini, 2022), it sorts the fifty states by their joint path from 2000 to 2022 on two series at once, democracy and the logged Black incarceration rate, and the BIC selects five well-separated classes, with an average assignment probability near 0.99 (Figure 5).

State trajectory classes (gbmt), 2000-2022: $ng = 5$

Class-mean paths. States typed jointly on democracy + log Black incarceration; ratio shown for the equity read.

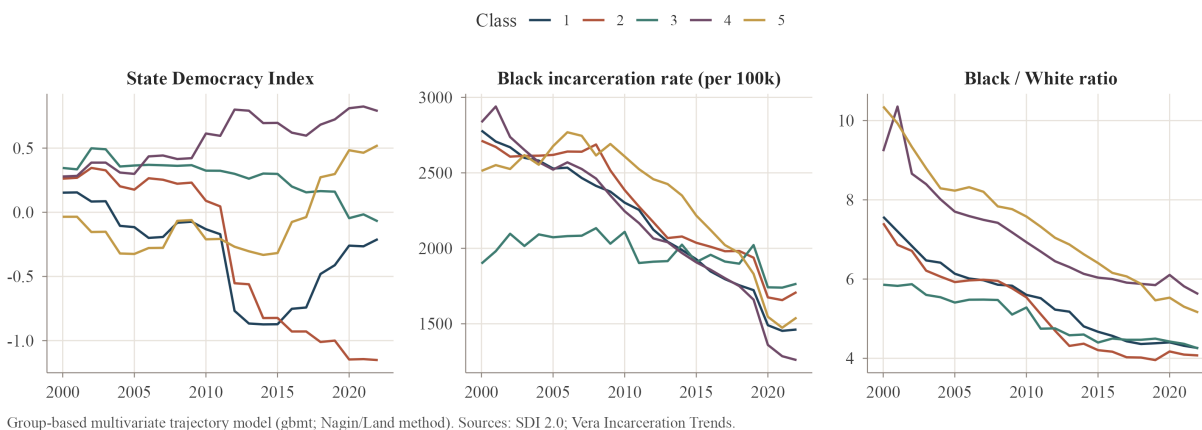


Figure 5: Five state trajectory classes from the gbmt model. Each line is a class-mean path for democracy, Black incarceration, and the Black/White ratio.

The classes arrange themselves along a single spectrum. States that grew more democratic decarcerated the most, while those that backslid the furthest, the group Grumbach (2023) calls the laboratories of backsliding, reduced Black incarceration the least and even saw their white rate climb. This is the same relationship as before, now drawn as a contrast between recognizable groups of states. The method invites overreading, though, and resisting that is the lesson here. Every class decarcerated, because the national decline runs through all of them, so the groups differ in degree rather than in direction. The ratio converged toward four or five across the board. And the BIC keeps improving as classes are added, so the figure of five is not firmly pinned down; highly autocorrelated series like these can produce classes that are not really there. The classes are useful summaries, not natural kinds, and they remain entangled with region. A method can hand back a clean typology that looks more solid than the data warrant, and choosing to read it as description rather than discovery is itself part of the analysis.

Demonstration 4: a reform by its text, or by its effect

The state-level patterns are associations, not causes, and to get closer to cause I turn to a natural experiment. In 2018 Florida passed Amendment 4, restoring the vote to about 1.4 million people with felony convictions, the largest such restoration in modern U.S. history. With only one treated state the appropriate tool is synthetic control (Abadie, Diamond, & Hainmueller, 2010), implemented with the tidysynth package (Dunford, 2023). The method assembles a synthetic Florida from a weighted combination of states that did not change their felon-voting laws, chosen so that the combination reproduces Florida's turnout history before the reform, and then reads the reform's effect as the gap that opens afterward. The outcome is VAP turnout, whose denominator does not move when people are re-enfranchised, and a placebo test provides inference by asking how unusual Florida's gap is against the same gaps computed for every donor state.

Florida vs. synthetic Florida: VAP turnout, 2000-2022

Amendment 4 effective 2019 (first election 2020); SB7066 reimposed fines-first mid-2019

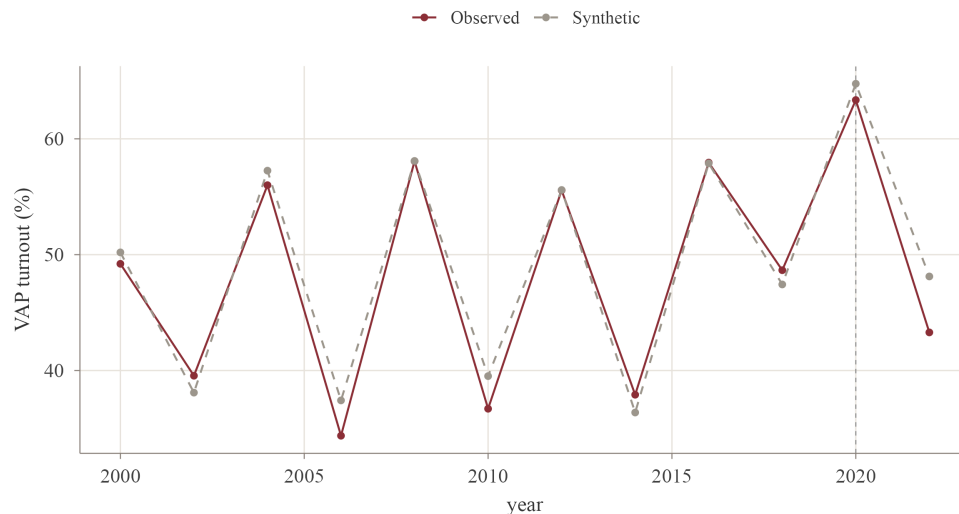


Figure 6: Florida compared with synthetic Florida on VAP turnout. The observed series (grey) tracks the synthetic series (pink) before and after the reform.

Placebo test: Florida's turnout gap vs. clean donor states

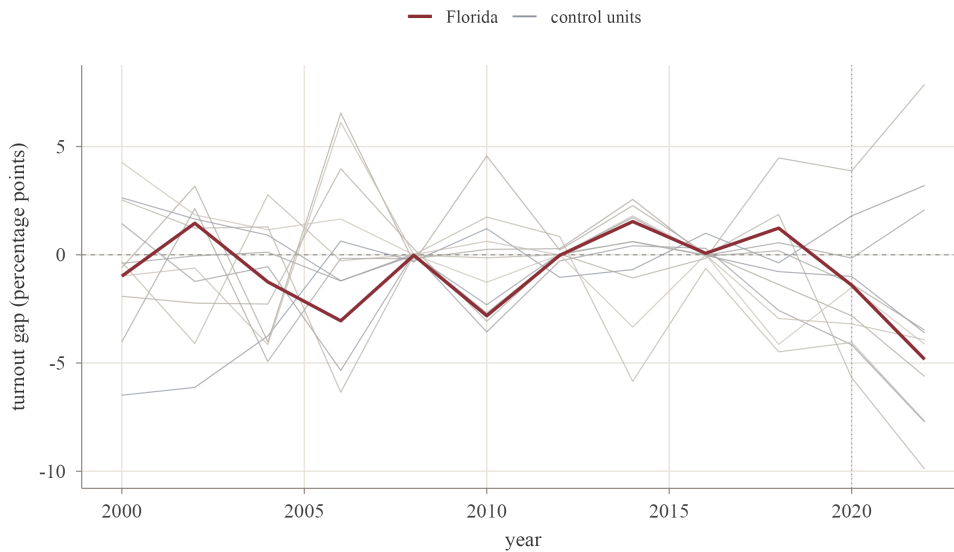


Figure 7: Placebo test. Florida's post-treatment gap (pink) sits inside the cloud of donor-state placebos.

The pre-reform fit is close, with a 2016 gap of only 0.07 points, which lends the comparison credibility (Figure 6). After the reform, the gaps are -1.4 points in 2020 and -4.8 in 2022, both negative rather than positive, and the placebo test returns a p-value of about 0.14, meaning Florida is not unusual among the donor states (Figure 7). Amendment 4 produced no measurable increase in turnout.

Two further checks indicate the null is robust rather than an artifact of the specification. An in-time placebo that moves the treatment back to 2016, before the reform, yields only small gaps in the genuinely pre-treatment years, about $+1.0$ points in 2016 and $+2.1$ in 2018; these are modest and positive, the opposite sign of the post-reform gaps, so the design does not manufacture the result, though the 2018 gap signals a mild pre-existing drift. A leave-one-out analysis that re-fits the model dropping each weighted donor in turn, with synthetic Florida drawing mainly on Arizona and New Hampshire, leaves the 2020 gap between -1.6 and $+1.1$ points and the 2022 gap between

−5.4 and +1.3, straddling zero in every case, so no single donor drives the estimate. The negative 2022 gap is therefore best read as a fluctuation around zero rather than as turnout suppression: it is not statistically distinguishable from zero and reverses when individual donors are removed.

A null result is only worth reporting if its source is visible, and here it is. The reform barely took effect. Florida's disenfranchised share fell only from 1.25 percent of the voting-age population in 2018 to 1.16 percent in 2020 and 1.14 percent in 2022, a nearly flat line rather than the steep drop that re-enfranchising 1.4 million people would imply. The explanation is SB7066, which required people to pay all outstanding court fines and fees before regaining the vote, a requirement that works as a wealth test on the franchise and falls hardest on poor and disproportionately Black citizens. By its text the reform enfranchised more than a million people; by its effect it enfranchised few. Which description counts as true depends on whether one reads the statute or the turnout, and here, once again, the framing is the finding.

Discussion

The four demonstrations make a single argument. In each, a routine decision settles the result: a measure of inequality reverses the sign, a comparison design shrinks a striking effect to a small one, a clustering method offers more structure than the data hold, and a reform reads as sweeping or negligible depending on whether one consults its text or its effect. The substantive conclusions matter, but they are secondary to this. On questions like these, measurement, design, and framing are not technicalities to be settled before the real work begins; they are where the answer is actually made. And the demonstrations need not stay separate. Fitting one Bayesian model of the underlying rates, as the middle section did, recovers the competing measures as contrasts of a single posterior;

it shows why the choices matter and lets the reversal be reported as a probability rather than a juxtaposition of figures.

That measurement and framing carry this weight is, as the background made clear, a well-worn point in the literature. What is easy to lose, stated only in the abstract, is how concretely and how quickly the choices bite, and how they accumulate. Here they are shown one after another, with numbers, on a case where the stakes are real: the most-cited measure of racial disparity in incarceration points the wrong way for a mechanical reason, and a celebrated voting-rights reform reads as a triumph or a non-event depending on the frame. For anyone who studies inequality, the practical counsel is cheap to follow. Carry more than one measure. Separate within from between. Treat data-driven groups as descriptions rather than kinds. Judge a policy by its effect rather than its text. And when the measures disagree, read the disagreement as a finding rather than a nuisance. Stated plainly, the substantive takeaways are modest. State democracy and racial inequality in punishment are linked only weakly, and only within states over time; the link is small beside a national decline in incarceration; and re-enfranchisement on paper can be undone by a fee. None of these is a breakthrough. Taken together, though, they show how easily the convenient answer turns out to be wrong.

Several limits bound the claims. Aggregate turnout is a blunt instrument: even a fully effective Amendment 4, involving about 8 percent of the voting-age population, most of whom never register or vote, might move statewide turnout by less than a point, below what this design can detect, so the null means there was no large aggregate effect rather than that re-enfranchisement is pointless, and a sharper test would follow the formerly incarcerated individually in voter files. The first three demonstrations are observational rather than experimental, and they support the framing of

the question rather than a causal answer to it. State democracy and incarceration are themselves estimated quantities. And the synthetic control rests on a single treated state, an imperfect midterm pre-fit, a small pool of clean donor states, and a 2020 treatment year disrupted by the pandemic.

Conclusion

This paper has used one case, race and democracy and punishment in the U.S. states, to make a methodological point: that the answer depends on how we measure and frame the question, often more than on the data. Change the measure and a disparity reverses; change the comparison and an effect shrinks; change the frame and a reform that freed a million voters did almost nothing. The lesson is not that the data are useless, but that the choices made around them deserve as much scrutiny as the findings themselves. On the questions of who is punished and who may vote, that scrutiny is not optional.

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